Homework #1 — Data Basics and EDA (15 pts)

MATH 240 — Introduction to Statistics, Fall 2026

Due Date: Friday, 9/11 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as a .pdf format. Any submissions not in this format are subjected to point deductions.
- Responses can be hand-written or typed. You may also use a knitted Quarto file to complete the assignment.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- For all questions involving code, please provide your code along with the output.
- Refer to syllabus for late work deductions.

Background

The first homework will focus on parts of a data frame and experimental design. Additionally, there will be an exploratory question that utilizes R in order to explore various components of the data through data visualizations and tables.

Part I: Textbook Problems (12 pts)

Complete the following problems:

1. Exercises from Chapter 1: #6, #16
2. Exercises from Chapter 2: #8, #12, #26

Part II: Palmer Penguins (3 pts)

Over the next few homework assignments, we will explore a dataset consisting of penguins! Specifically, we will be looking at various penguin species observed on three islands in the Palmer Archipelago, Antarctica.

We will be using R to work with the data. For this question, you will turn in a separate, knitted Quarto document with code and output related to a brief exploratory data analysis we will complete with this dataset. Follow all instructions below.

1. Use the following code **in your CONSOLE** to install the `palmerpenguins` package. You do not need to include this code in the actual Quarto document you will turn in.
   - `install.packages("palmerpenguins")`
2. Create a new Quarto document. Title the document "HW1, Part II" and add your name in the Name box.
3. Use the following code to bring the `penguins` dataset into your R environment:
   - `library(palmerpenguins)`
   - `library(dplyr)`
   - `library(ggplot2)`
   - `data(penguins)`
4. Exploring the `penguins` dataset, gather the following items using R code:
   (a) Number of observations and variables
   (b) Frequency table of penguin species
   (c) An appropriate plot of only the penguins' sex (NAs are fine to keep in there)
5. Go to the introductory article of this dataset here. Pick one plot (with its code) and replicate it in R, but with at least TWO adjustment to the code. You can change visual components, variables, etc.
   (a) Show the adjusted plot in the document.
   (b) Write 2-3 sentences describing what you changed, and how that affected the plot. Additionally, discuss what you can hypothesize about the information seen in your plot, using visual cues.